

MATH0104 Modular Forms

<i>Year:</i>	2026–2027
<i>Code:</i>	MATH0104
<i>Level:</i>	7 (UG), 7 (PG)
<i>Normal student group(s):</i>	UG: Y3, Y4 Mathematics programmes PG: MSc Mathematics
<i>Value:</i>	15 credits (7.5 ECTS credits)
<i>Term:</i>	1
<i>Assessment:</i>	70% examination, 30% departmental test(s)
<i>Normal Pre-requisites:</i>	MATH0013 (Analysis 3: Complex Analysis) MATH0014 (Algebra 3: Further Linear Algebra)
<i>Lecturer:</i>	Dr O Gregory

Course Description and Objectives

This module aims to offer an overview of the basic notions that appear in the classical theory of modular forms. These are analytic objects encoding a lot of arithmetic information which makes them a central point of study in number theory and arithmetic geometry. Modular forms also arise naturally in a variety of other research fields like transcendence proofs, differential equations and mirror symmetry.

The main objects of interest are functions on the complex upper half plane that transform in a special way under the action of $SL_2(\mathbb{Z})$. Two concrete constructions of such functions will be covered: Eisenstein series and theta series. We will also show that the space of modular forms for a specific weight is finite dimensional, which makes all these functions algorithmically computable. Furthermore, we will introduce Hecke operators and show that the L-functions associated to eigen functions are of arithmetic nature. Throughout the module, several applications to congruences and positive definite quadratic forms will be highlighted.

Recommended Texts

- LJP Kilford, *Modular Forms: A Classical and Computational Introduction* (Imperial)
- JP Serre, *A Course in Arithmetic* (Springer)
- W Stein, *Modular Forms, a Computational Approach* (AMS)
- H Cohen and F Stromberg, *Modular Forms: A Classical Approach* (AMS)

Detailed Syllabus

- The modular group
- Modular functions and modular forms for the full modular group
- The dimension of the space of modular forms of weight k
- Hecke operators
- The Petersson Inner Product.
- The L-function of a modular form

- Theta functions
- Applications to quadratic forms

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